

Enterprise GenAI Case Study

Multi-Model LLM Architecture for Payment Sanctions

Cost Comparison

Before (Single-Model)

Monthly Inference	\$600,000
Manual Review Cost	\$60,000,000
False Positive Rate	40%
Review Time	2-3 days

After (Multi-Model)

Monthly Inference	\$79,800
Manual Review Cost	\$23,400,000
False Positive Rate	15.6%
Review Time	2-4 hours

Impact Metrics

\$35M

Monthly Savings

61%

False Positive
Reduction

90%

Faster Review

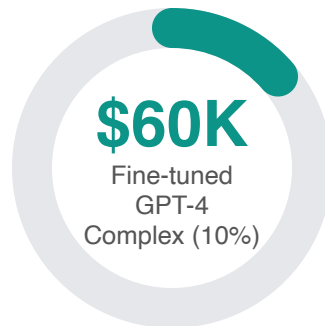
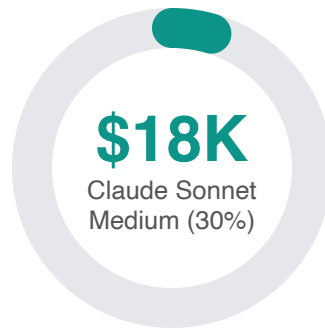
\$2B+

Revenue Enabled

Cost Breakdown by Model

\$1.8K

GPT-4o-mini
Simple (60%)



Implementation Journey

1

Phase 1: Foundation (Months 1-3)

Established AI governance framework, formed AI Ethics Committee, defined data privacy protocols, and created model evaluation criteria.

2

Phase 2: Prototype (Months 4-6)

Built proof of concept with single model, established accuracy benchmarks, created cost analysis baseline, and delivered stakeholder demos.

3

Phase 3: Multi-Model (Months 7-9)

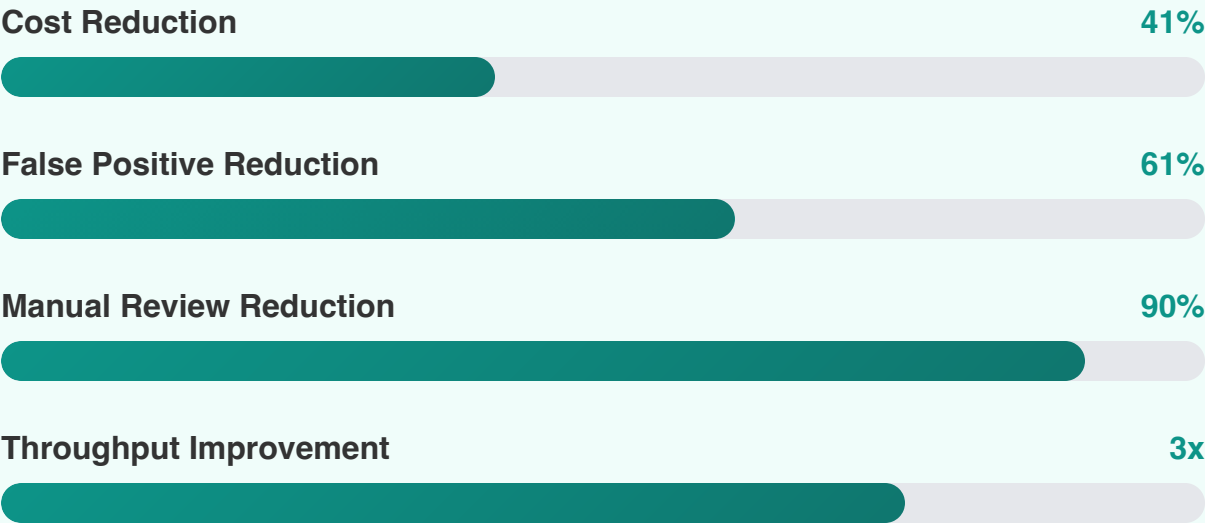
4

Implemented intelligent routing, integrated multiple LLM providers, built cost optimization layer, and created A/B testing framework.

Phase 4: Agentic Workflows (Months 10-12)

Deployed production-ready multi-agent applications, built agent orchestration system, implemented workflow automation, and created monitoring dashboards.

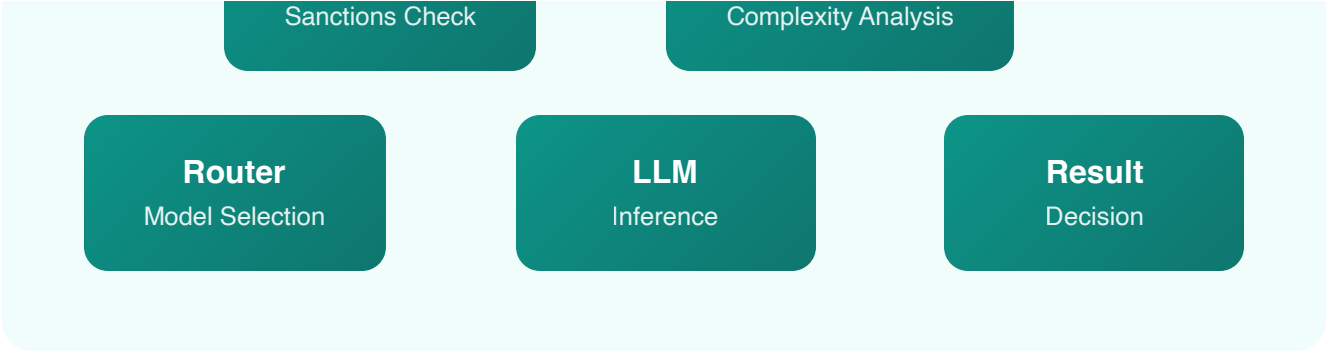
Results Breakdown



Multi-Model Architecture Flow

Request

Classifier



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